

Indoor air can be up to 5× more toxic than outdoor air and you can't see it.

Invisible spikes in CO₂ and toxic particulates build up in homes and offices every day, unnoticed. **PoWear** is a wearable safety device that senses hazardous air in real time and shows users exactly where to act, before it affects their family's health, with zero cameras and zero microphones.

89.1M

REAL-WORLD SENSOR READINGS
POWERING THE DETECTION MODEL

77%

FASTER REDUCTION OF HAZARDOUS CO₂
LEVELS

≤34μs

ON-DEVICE HAZARD ALERT LATENCY

0

CAMERAS OR MICROPHONES, 100%
PRIVATE

STEP 01 · SENSE

Detection Engine

Trained on 89.1M readings from 30 real homes over 13,646 hours, PoWear knows exactly what a dangerous CO₂ or particulate spike looks like before it becomes harmful.

STEP 02 · SEE IT

PoWear Wearable

A wristband that turns invisible pollution into a glowing AR bubble users can point a fan at, clearing a danger zone in seconds, not hours. 3.8× faster hotspot clearance in testing.

STEP 03 · STAY SAFE

On-Device AI

Learns a household's risky habits: frying, poor ventilation, overnight buildup from air chemistry alone, and warns before they recur. No cloud, no privacy risk.

MISSION

"People spend nearly 90% of their lives indoors, trusting air they can't see is safe. PoWear exists to break that false sense of security — turning invisible risk into a warning you can act on, instantly and privately."

— DaltonAir Team, UbiNet Lab, IIT Kharagpur

User preference for AR alerts vs. plain app	84.3%
Cooking / activity recognition precision	99.68%

WHY WE WIN ON PRIVACY

SENSING MODALITY	PRIVACY RISK	PRECISION
Optical video cameras	Extreme	98.20%
Acoustic microphones	High	91.40%
Air Pollutants	0% (pervasive)	99.68%

Open dataset & research	github.com/prasenjit52282/dalton-dataset
Pre-orders opening	Early 2027

TRACTION & ROADMAP

2024

DALTON dataset published at NeurIPS — 89.1M readings, 30 homes

2026

AR interaction research accepted at CHI; On-Device AI accepted at DCOSS-IoT

Now

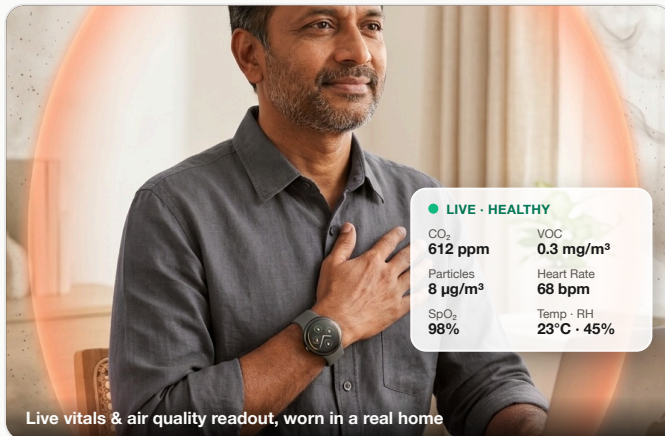
PoWear v2 hardware & on-device AI validation with biomarkers

Early 2027

Pre-orders open; waitlist & hiring for launch team underway

PoWear :

the wearable that senses hazardous air



Live vitals & air quality readout, worn in a real home